11992 - Discovery of the first quantum-cooling star

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	F322W2 imaging	NIRCam Imaging	(1) PSR-J2222-0137

ABSTRACT

White dwarfs crystallise as they cool. Reaching a low enough temperature, a white dwarf should, theoretically, reach its Debye temperature and then later begin the Debye cooling stage, where quantum effects start to dominate the lattice cooling. In this phase, the heat capacity drops as the cube of the temperature, causing the white dwarf's temperature to suddenly decrease very rapidly. Such 'black dwarfs' should be a significant class of cold stellar remnant in the Milky Way, but have never so far been found. The massive white dwarf that is the companion to the radio pulsar PSR J2222-0137 is to date the coldest white dwarf known. Its mass (1.32 Mo) and distance (268 pc) are extremely well-constrained from high-precision radio timing data. However, deep HST optical imaging has failed to detect it, placing the temperature at <3000 K. We propose here NIRCam imaging at 3.3 micron which, due to the low temperature of the source, will be much more sensitive than the HST imaging, reaching blackbody temperatures down to 900 K. To cool to temperatures significantly below 3000 K in the lifetime of the Galaxy, the source must have entered the Debye cooling regime. Detection of the WD would be an astonishing result, implying late heating via an unknown mechanism, and providing novel information on

the properties of exotic dense matter under these conditions, which are so far not well-understood. This observation would mark the first clear demonstration of a white dwarf clearly affected by Debye cooling, demonstrating the first astrophysical object in this quantum-cooling state of matter.

OBSERVING DESCRIPTION

We want to detect, or place sufficiently strong limits on the luminosity of, the coldest white dwarf currently known. The white dwarf is the companion to the binary radio pulsar PSR J2222-0137, and because of the high-precision timing of the radio data, its mass and distance are very well constrained, as well the orbital parameters of the binary. HST gri observations provide only upper limits, clearly suggestive of a white dwarf that has reached the Debye cooling stage. To test this, and constrain the Debye cooling regime for the first time, it is important to be able to observe the white dwarf close to its peak emission, for which NIRCAM is ideally suited for temperatures around 1000-2000 K. To make the most sensitive possible constraint, we request point source imaging in the broadband F332W2 filter. An observation of only 1.6 hours will reach a limiting temperature below 1000K, by far the lowest temperature ever reached for a white dwarf.

Proposal 11992 - Targets - Discovery of the first quantum-cooling star

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PSR-J2222-0137	RA: 22 22 6.0290 (335.5251208d) Dec: -01 37 16.04 (-1.62112d) Equinox: J2000	Proper Motion RA: 0.002801121139226317 sec of time/yr Proper Motion Dec: -0.01499999996212864 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Binary pulsars, Neutron stars, Pulsars, White dwarfs]
Extended=NO

Observation

Proposal 11992, Observation 1: F322W2 imaging

Fri Mar 13 22:05:52 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

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Template

Module	Subarray
B	FULLP

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	SUBARRAY_DITHER	3	SMALL-GRID-DITHER		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F150W2	F322W2	DEEP8	10	1	3	3	6055.538	146490